

v_o/i_s in
by the
2-k Ω

FIGURE P4-3

 **4-4 (a)** Find the voltage gain v_o/v_s and current gain i_o/i_x in Figure P4-4.

(b) Verify your answers by simulating the circuit in OrCAD.

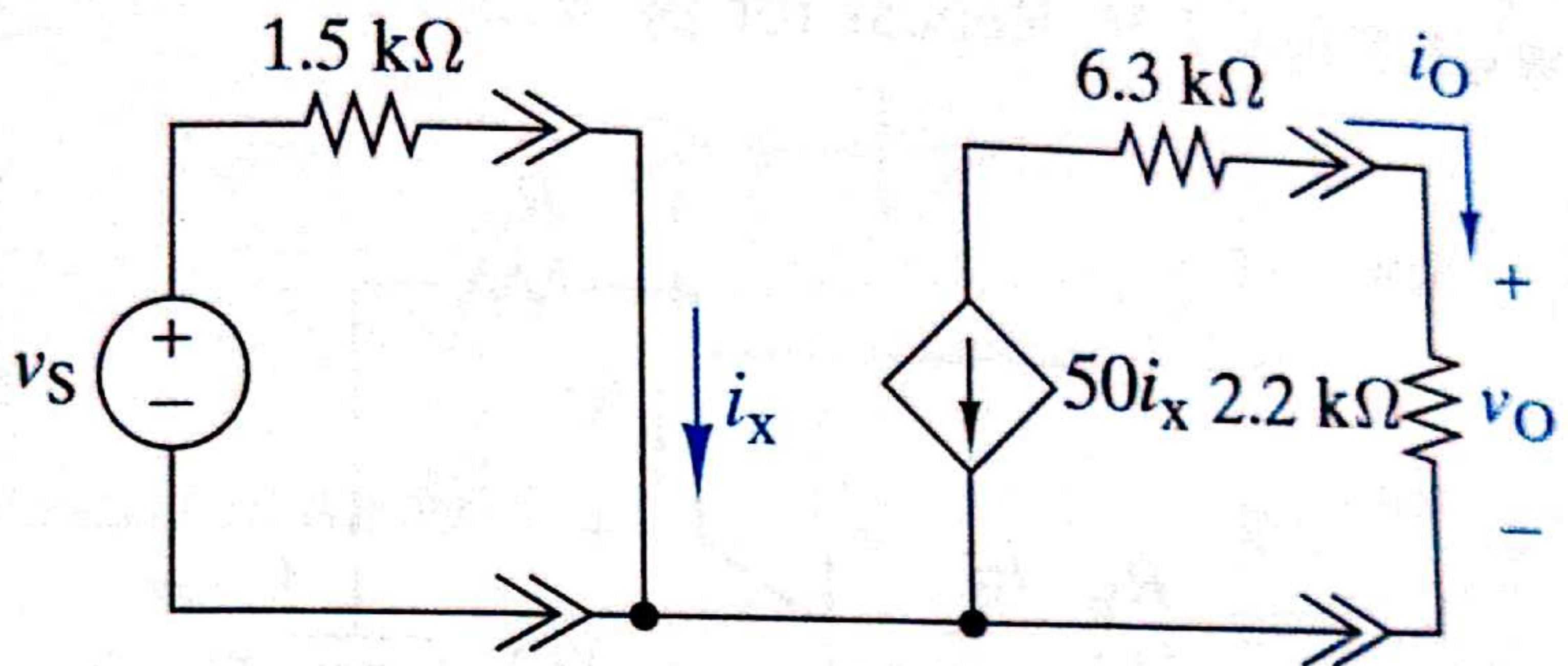


FIGURE P4-4

4-5 Find the voltage gain v_O/v_S in Figure P4-5.

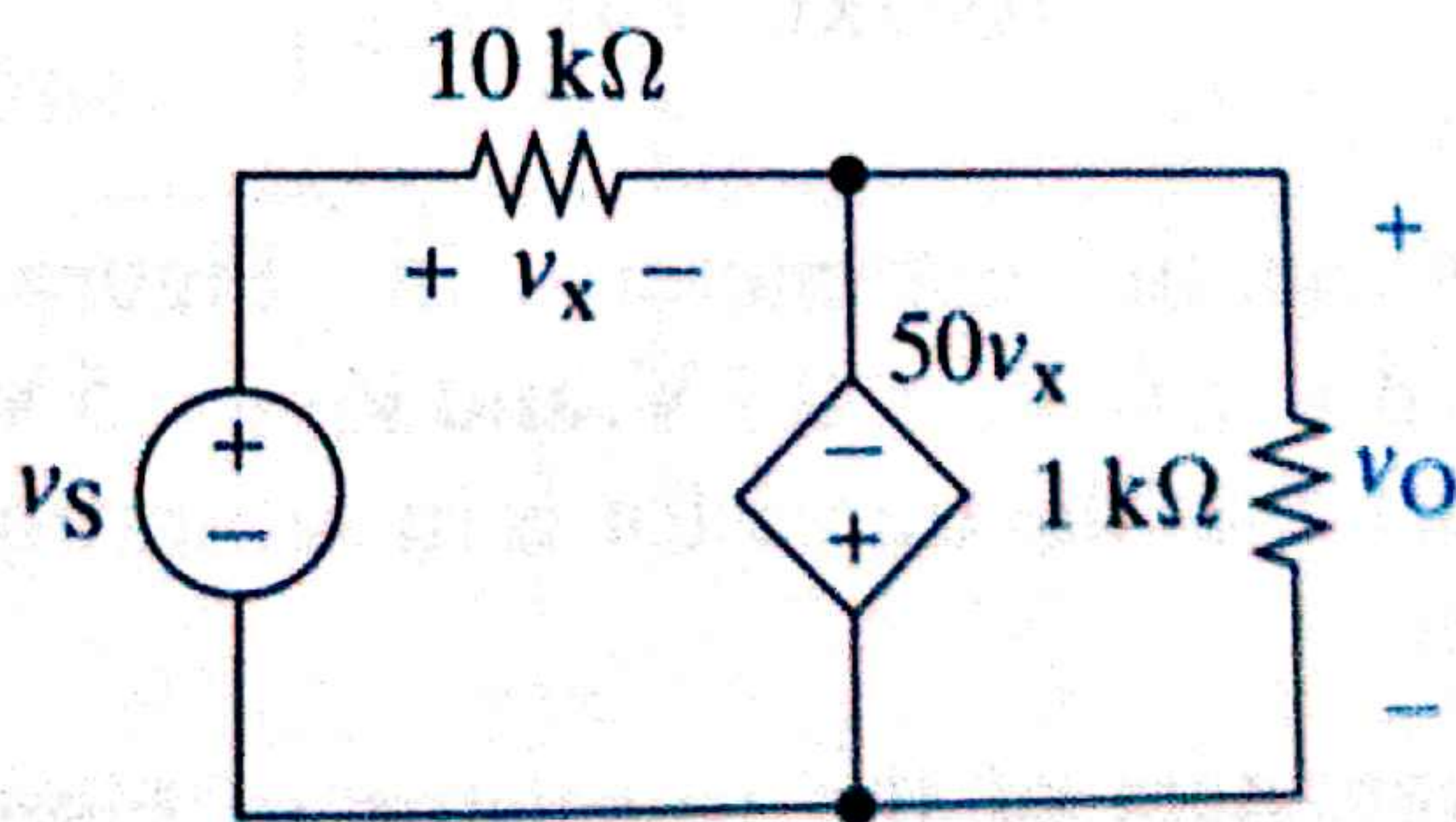


FIGURE P4-5

4-6 Find the voltage gain v_O/v_S in Figure

FIGURE P4-21

4-22 What is the range of the gain v_O/v_S in Figure P4-22?

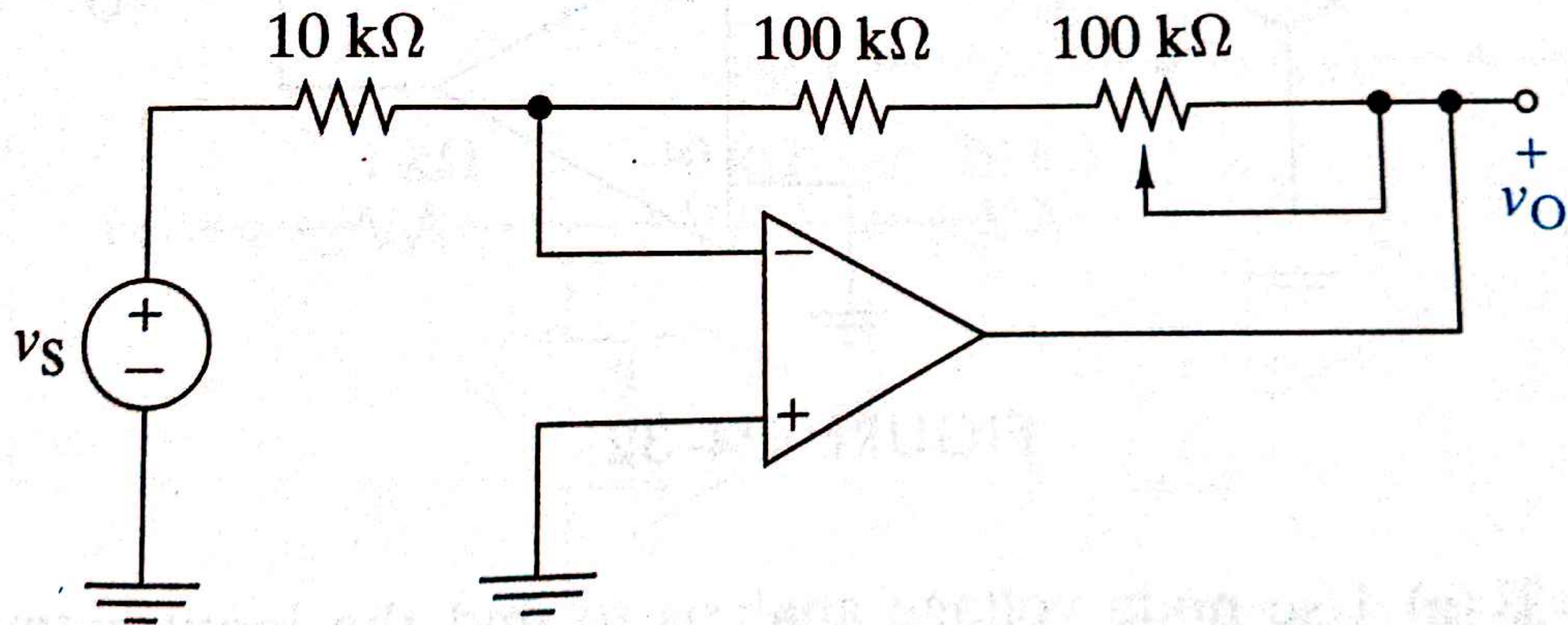


FIGURE P4-22

4-2
1

4–23 (a) Find v_O in terms of v_S in Figure P4–23.

(b) Find i_O for $v_S = 1$ V. Repeat for $v_S = 3$ V.

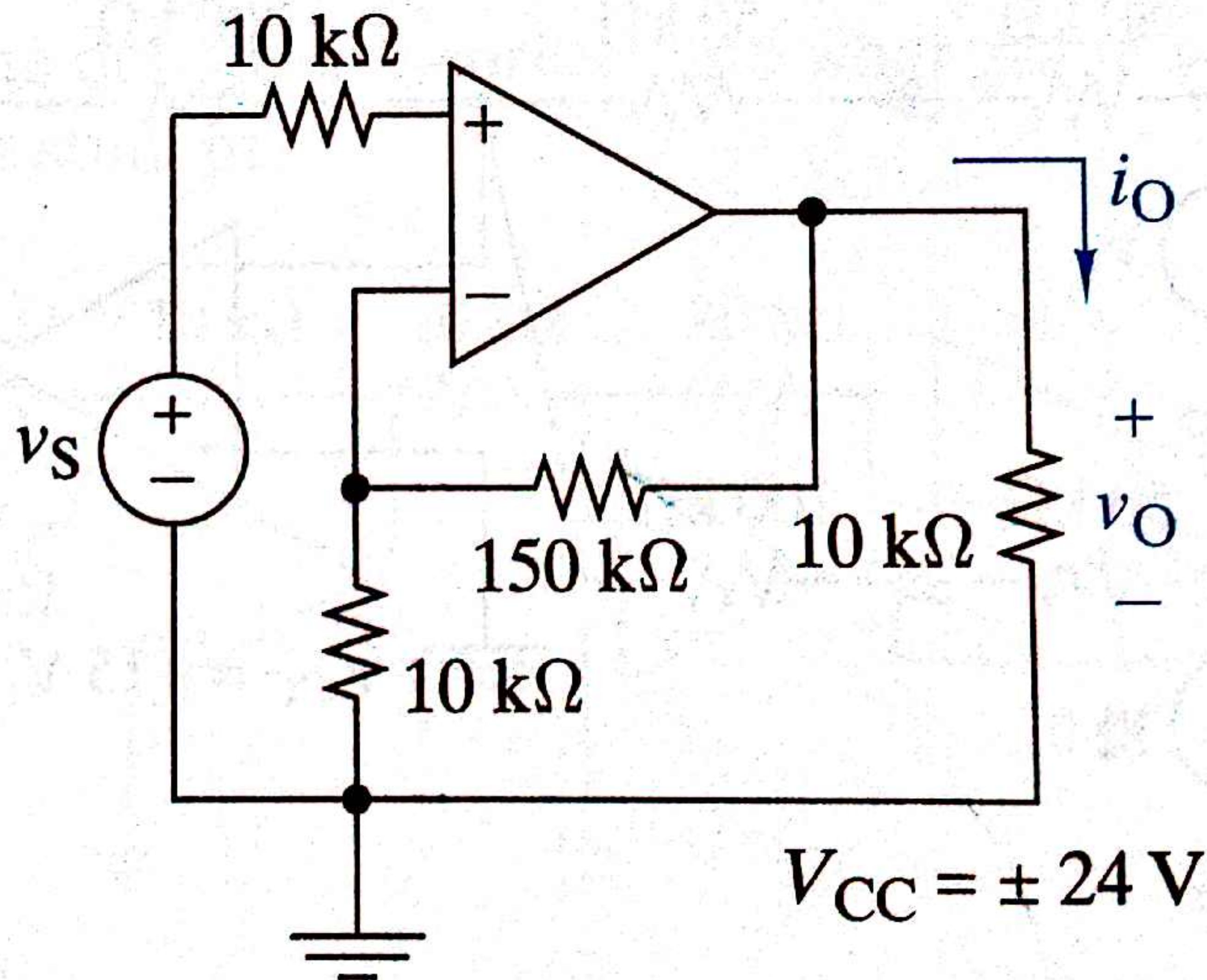


FIGURE P4–23

The resistance of the feedback resistor and values of the input resistors R_1 , R_2 , and R_3 .

4-28 Find v_O in terms of the inputs v_1 and v_2 in Figure P4-28.

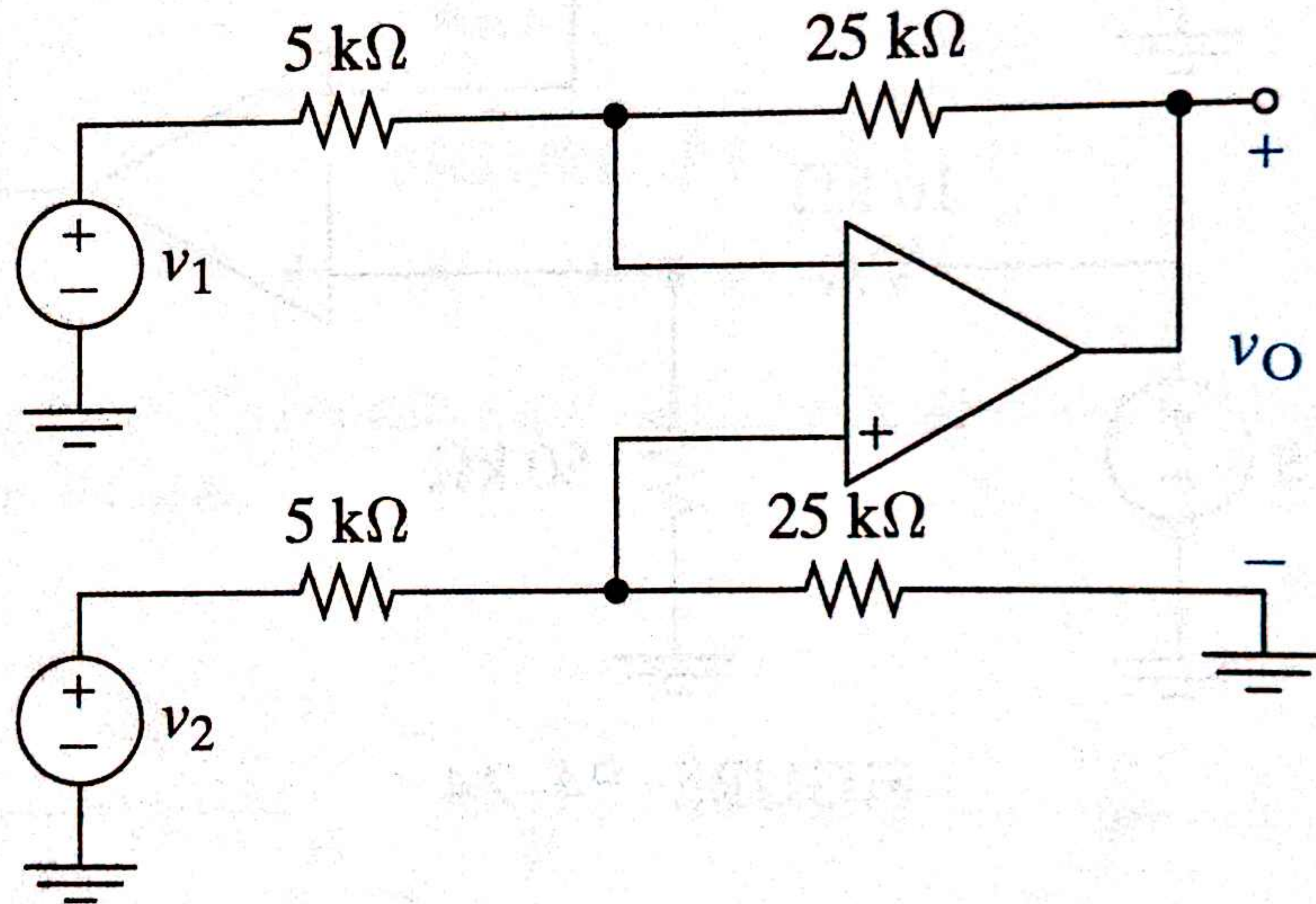


FIGURE P4-28

4-29 The switch in Figure P4-29 is open. Find v_O in terms of the inputs v_1 and v_2 .

4-34 Use node voltage analysis in Figure P4-34 to show that $i_O = -v_S/2R$ regardless of the load. That is, show that the circuit is a voltage-controlled current source.

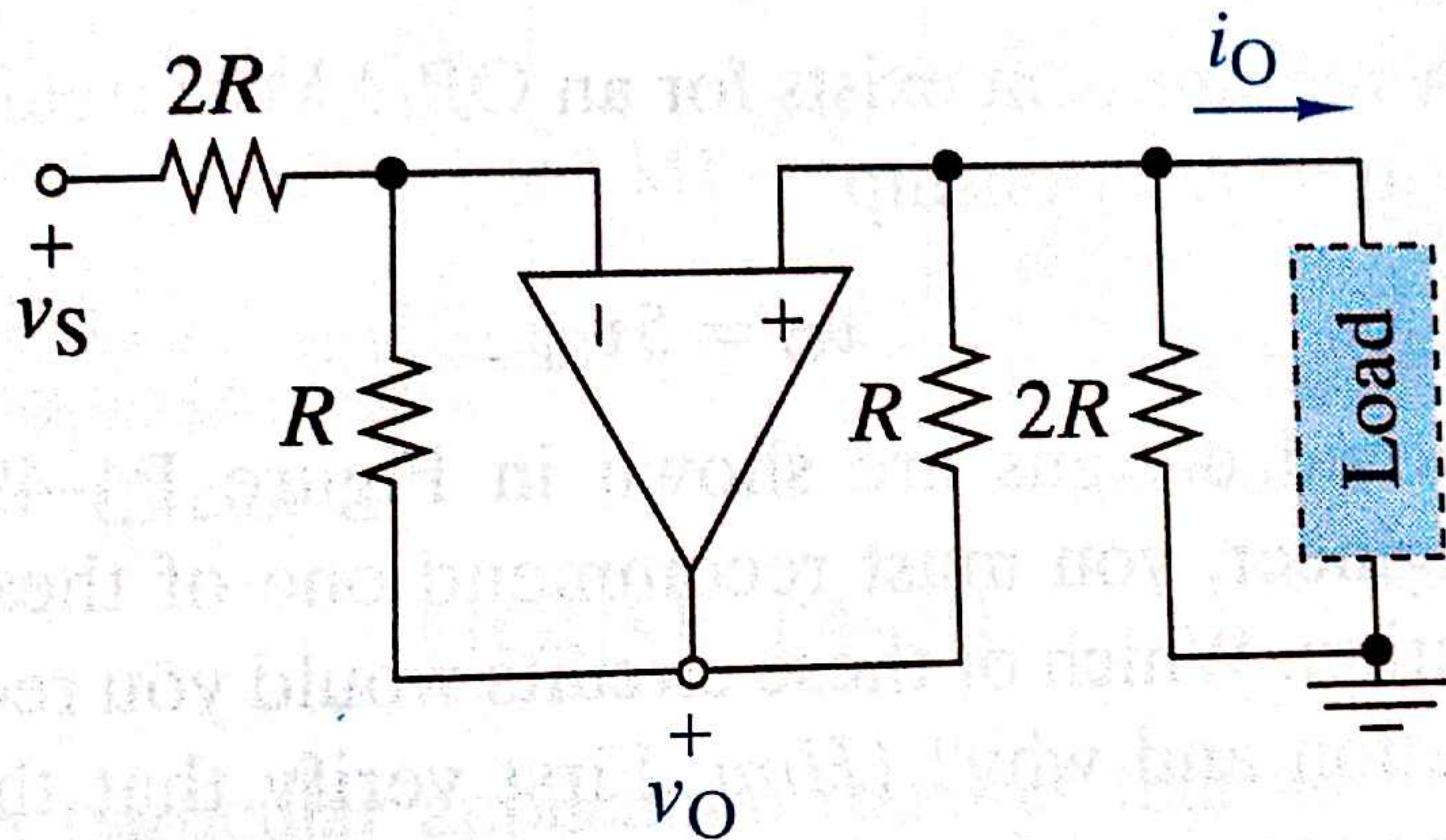


FIGURE P4-34

FIGURE P4-37

-35.

4-38 Find the output v_2 in terms of the input v_1 in Figure P4-38.

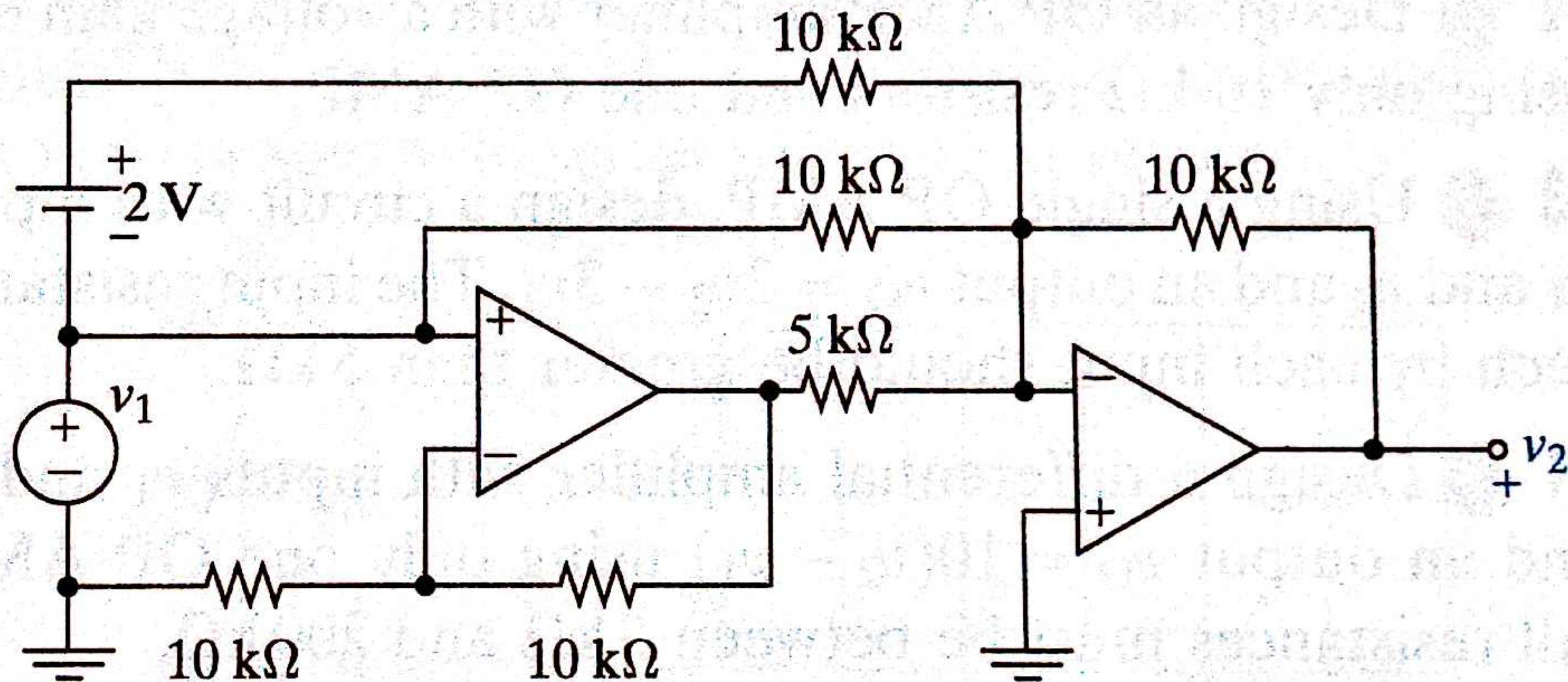


FIGURE P4-38

in

4-44  Design a differential amplifier with inputs v_1 and v_2 and an output $v_O = 10(v_2 - v_1)$ using only one OP AMP. All resistances must be between $5\text{ k}\Omega$ and $200\text{ k}\Omega$.

4-45  Using no more than two OP AMPs, design an OP AMP circuit with inputs v_1 , v_2 , and v_3 and an output $v_O = -3v_1 + 2v_2 - 5v_3$.

 4-46  Design the interface circuit in Figure P4-46 so that 10 mW is delivered to the $100\text{-}\Omega$ load. Repeat for a $100\text{ k}\Omega$

- 4-55 **D** A Chromel-Constantan thermocouple (curve E) has the characteristics shown in Figure P4-55. Design an interface that will produce a 0–5 V output where 0 V refers to 0°C and 5 V refers to 1000°C. The transducer can be modeled as a voltage source in series with a 15- Ω resistor.

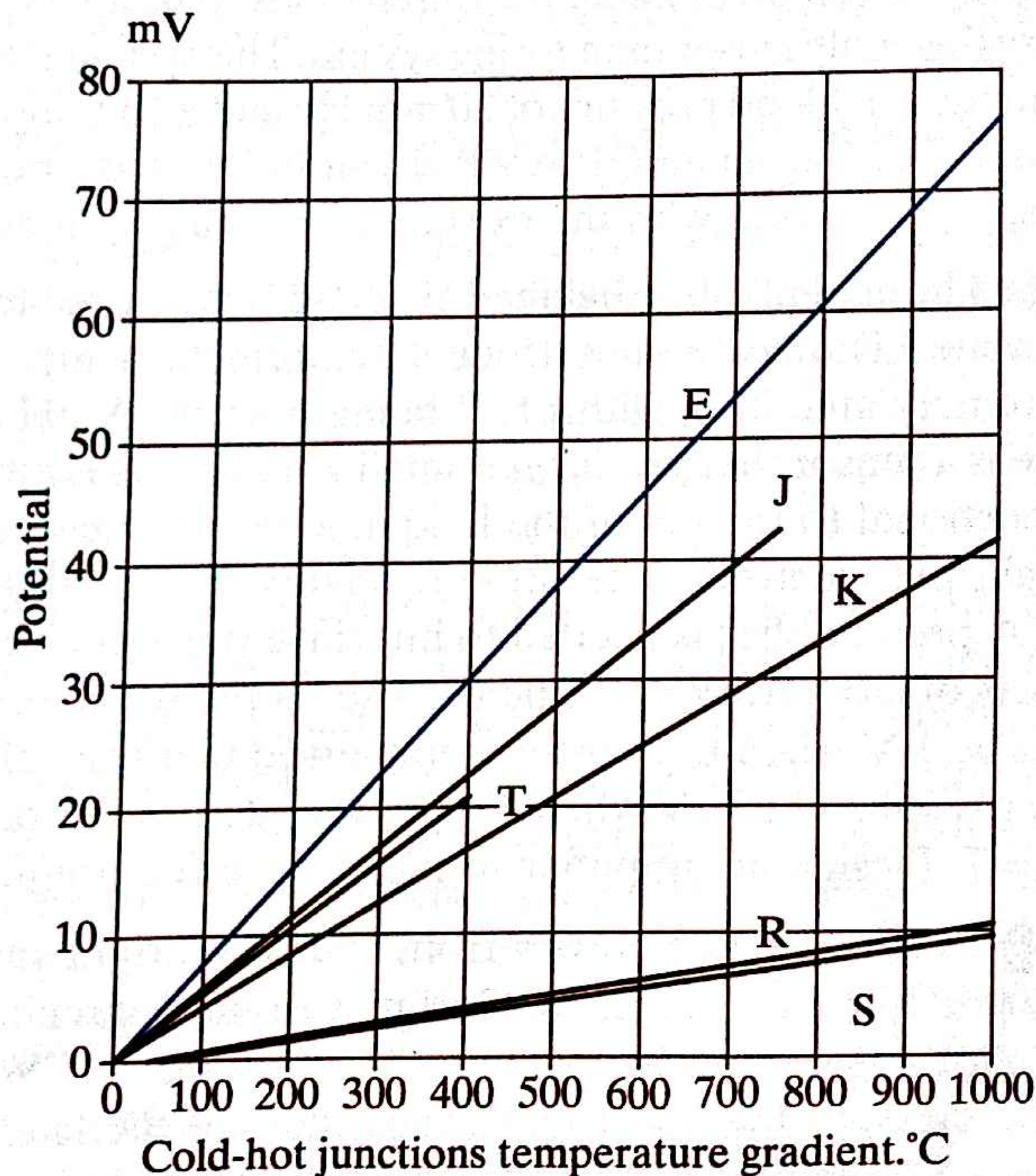


FIGURE P4-55

4-66 Thermometer Design Problem

There is a need to design a thermometer that can read temperatures from 350°C to 1100°C to monitor the temperature of a rocket's nose cone. The output will feed a 0–5 V ADC prior to transmission of the temperature data to the ground. A reading of 350°C will deliver 0 V to the ADC, while 1100°C will deliver 5 V. Select from Figure P4-55 an appropriate thermocouple and design the instrumentation amplifier. Other criteria are: that the transducers have a $15\text{-}\Omega$ series resistance; the fewer OP AMPs the better, since power is limited; the maximum OP AMP gain is 1000; and the bias voltage, if needed, must be from the available 5-V supply.

4-67 High Bias Design Problem

A high-pressure sensor is designed to operate under